



Solid electrolyte interphase formation behavior on well-defined carbon surfaces for Li-ion battery systems

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ABSTRACT

Solid electrolyte interphase (SEI) formation behavior was carefully tracked on the well-defined edge and basal surfaces of carbon nanofibers. The effects of boron doping on SEI formation were also examined. Platelet carbon nanofibers (PCNFs) successfully afforded well-defined edge and basal surfaces. Surface morphology and chemistry were controlled by graphitization, nitric acid treatment and boron doping. The surface morphologies of various well-defined PCNFs were observed by transmission electron microscopy (TEM) and X-ray photoelectron spectroscopy (XPS). The SEI of PCNF with edge surfaces was more abundant and four times thicker than those of PCNF-G and PCNF-B-G with basal surfaces. The SEI of PCNF-G-NA without boron doping was much more abundant and three times thicker than that of PCNF-B-G-NA with boron doping.

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1. Introduction

A long-standing goal in battery research is to understand the mechanisms of aging and degradation of lithium-ion (Li-ion) batteries. This is important for Li-ion batteries for transportation applications, where battery lifetimes of 10–15 years are required.

An understanding of the solid electrolyte interphase (SEI) is essential because it affects the characteristics of Li-ion battery systems, including the life cycle, shelf life, rate capability, and safety [1]. The SEI formation takes place during the first few cycles and the SEI is responsible for irreversible capacity loss. Although the SEI acts as a solid-state diffusion path for lithium ions as well as a passivation layer against co-intercalation of solvent molecules [2], excessive SEI formation is quite possible to limit cathode capacity.

The SEI is usually composed of reduction products of salt anions, such as LiF, LiCl, and Li₂O, and decomposition products of the electrolyte solvent, such as lithium alkyl carbonates and lithium carbonate [3,4]. Many researchers have shown that the surface chemistry and morphology of the SEI differ markedly according to the state of the carbon surfaces, e.g., the cross-section and basal planes of highly ordered pyrolytic graphite (HOPG) and types of

carbonaceous materials, such as soft carbon and hard carbon [5–7]. The SEI formed on the basal plane of HOPG is rich in organic compounds due to preferential reduction of solvents, whereas the SEI on the cross-section is rich in inorganic compounds due to preferential reduction of salt anions. The SEI on the basal plane was reported to be 3–5 times thinner than that on the cross-section. The SEI on hard carbon was also reported to be thicker than that on soft carbon. The SEI content on soft carbon was similar to that on the basal plane of HOPG, whereas SEI composition on hard carbon was similar to that on the cross-section of HOPG. Fujimoto et al. showed that boron doping increased the initial coulombic efficiency of graphite prepared from naphthalene-based pitch regardless of the type of boron compound present [8].

In our previous work, we showed that the ideal edge and basal surfaces were very difficult to obtain through simple mechanical polishing of HOPG and that the polished edge surfaces of protruding graphite layers were exposed to the experimental environment, especially to electrolyte molecules [9]. It was also found that the well-defined edge and basal surfaces of platelet carbon nanofibers (PCNFs) could be controlled by high-temperature heat treatment and subsequent nitric acid treatment [10]. These observations suggest that carbon nanofibers could be used as a model material to investigate the SEI and the state of the surface properties of graphite anodes in Li-ion batteries.

The present study was performed to track the SEI formation behavior on well-defined edge and basal planes of PCNF by transmission electron microscopy (TEM) and to study the effects of its boron-doped surfaces on SEI formation.

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Table 1
Physical properties of as-prepared, graphitized, nitric acid treated, and boron-doped PCNF.

Samples	Elemental compositions (wt%)							Crystallographic parameters		N ₂ -BET	Raman spectra
	H	C	N	O _{diff} ^f	H/C ratio (%)	O _{diff} /C ratio (%)	Ash	d ₀₀₂ (Å)	Lc ₀₀₂ (nm)	Surface area (m ² g ⁻¹)	R value (I _D /I _G)
PCNF ^a	0.21	99.16	0.00	0.49	0.21	0.49	0.14	3.365	57	98	1.10
PCNF-G ^b	0.10	99.87	0.00	0.03	0.10	0.03	0.00	3.366	62	65	0.52
PCNF-G-NA ^c	0.16	98.15	0.09	1.60	0.16	1.63	0.00	3.364	73	66	0.85
PCNF-B-G ^d	0.04	96.52	0.01	3.43	0.04	3.55	0.00	3.359	99	48	0.74
PCNF-B-G-NA ^e	0.23	95.48	0.06	4.23	0.24	4.43	0.00	3.359	>100	45	0.88

^a As-prepared platelet carbon nanofibers (PCNFs).

^b Graphitized PCNF.

^c Graphitized and nitric acid treated PCNF.

^d Boron-doped and graphitized PCNF.

^e Boron-doped, graphitized, and nitric acid treated PCNF.

^f The assays of O (O_{diff}) were defined by subtracting the sum of the contents of H (hydrogen), C (carbon), N (nitrogen), and ash from 100%.

2. Experimental

2.1. Preparation of PCNF and introduction of specific surfaces

PCNFs were prepared by catalytic decomposition of carbon monoxide and hydrogen gases on an iron catalyst at 640 °C for 4 h as described previously [10,11]. To remove the iron catalyst, synthesized platelet-type CNF were treated with 10 wt% HCl for 72 h, filtered with distilled water, and then dried at 110 °C for 12 h under vacuum (PCNF). PCNF was heat treated at 2800 °C for 10 min for further graphitization (PCNF-G), treated with 10 wt% HNO₃ at 155 °C for 28 h (PCNF-G-NA) using reflux condenser with oil bath under atmospheric pressure. The 5 wt% boron doping into PCNF was carried out by ball milling PCNF with boric acid (99.99% trace metals basis; Sigma-Aldrich) and continuous heat treatment. The PCNF, boric acid, and alumina balls (10 mm in diameter) were placed into a bottle with a ball-to-weight ratio of 22.5. Ball milling was performed on a V-IML planetary mill at a rotation speed of 140 rpm for 24 h. The ball-milled mixture was heat treated at 2800 °C for 10 min for further graphitization (PCNF-B-G) and then treated with 10 wt% HNO₃ at 155 °C for 28 h (PCNF-B-G-NA).

2.2. Characterization

Carbon, hydrogen, and nitrogen contents of the samples were measured by elemental analysis. The assays of O (O_{diff}) were defined by subtracting the sum of the contents of C, H, N, and ash from 100%. The crystallographic parameters (d₀₀₂ and Lc₀₀₂) of the CNF were measured by X-ray diffraction (XRD) (Ultima-III, Cu Kα; Rigaku). The 15 wt% silicon (99.999%; Soekawa Chemicals) was added to the CNF as a standard material to calculate the crystallographic parameters of the CNF according to the JSPS procedure [12]. Surface area was measured by BET analysis of N₂ adsorption isotherms at 77 K using a specific surface area analyzer (Belsorp-max S; Nippon Bell). The degree of graphitization was measured by Raman spectroscopy (NRS 3000; JASCO) with a 532-nm Ar laser as the excitation source. The optical power at the sample position was maintained at 12 mW. The I_D/I_G ratio (R value), the intensity ratio of the 1355 cm⁻¹ band (D band) and the 1580 cm⁻¹ band (G band), is determined from Raman spectra. Structure and surface morphology of the CNF were examined by scanning electron microscopy (SEM, JSM-6700F; JEOL) and TEM (200 kV, JEM-2100F; JEOL). The electrochemical properties of the CNF were determined by charge (Li⁺ intercalation)/discharge (Li⁺ de-intercalation) measurements using a CR2032 two-electrode button cell. Li metal foil was used as the counter electrode and reference electrode. The working electrode was prepared by mixing 90 wt% active material and 10 wt% polyvinylidene fluoride (PVDF) dissolved in *N*-methylpyrrolidone

(NMP). The mixed slurry was coated onto copper foil and then dried in a vacuum oven at 120 °C for 3 h. The electrolyte solution was 1 M LiClO₄ dissolved in ethylene carbonate (EC)/diethyl carbonate (DEC) (1:1 vol%; Kishida Chemical). Charge/discharge was performed at a current density of 372 mA g⁻¹ between 0.003 and 3 V vs. Li/Li⁺ (Toscat-3100; Toyo System). The specimens were scraped from the copper, washed in dimethyl carbonate (DMC, Kishida Chemical) and then dispersed on copper grids to observe the SEI film by TEM. Assembly of cells was carried out at room temperature in an argon-filled glove box with a dew point less than -70 °C. X-ray photoelectron spectroscopy (XPS) measurements were carried out on a JPS-9000MC (JEOL) with monochromatic Mg Kα radiation (1253 eV) under 7 × 10⁻⁷ Pa. The depth profile was determined by 0.5 kV argon-ion sputtering.

3. Results and discussion

3.1. Physical properties of PCNF

Table 1 shows the elemental compositions, crystallographic parameters, surface area changes and I_D/I_G ratio (R value) of as-prepared, graphitized, nitric acid treated, and boron-doped PCNF.

Heat treatment at 2800 °C increased the carbon content and decreased the oxygen content of PCNF; however, nitric acid

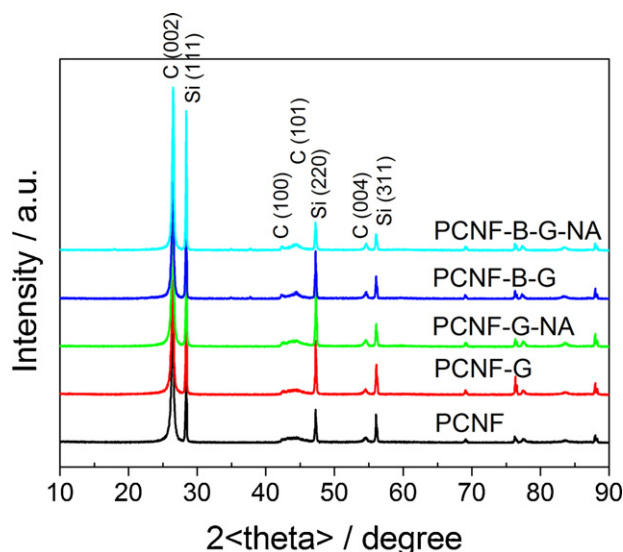


Fig. 1. XRD profiles of as-prepared, graphitized, nitric acid treated, and boron-doped PCNF.

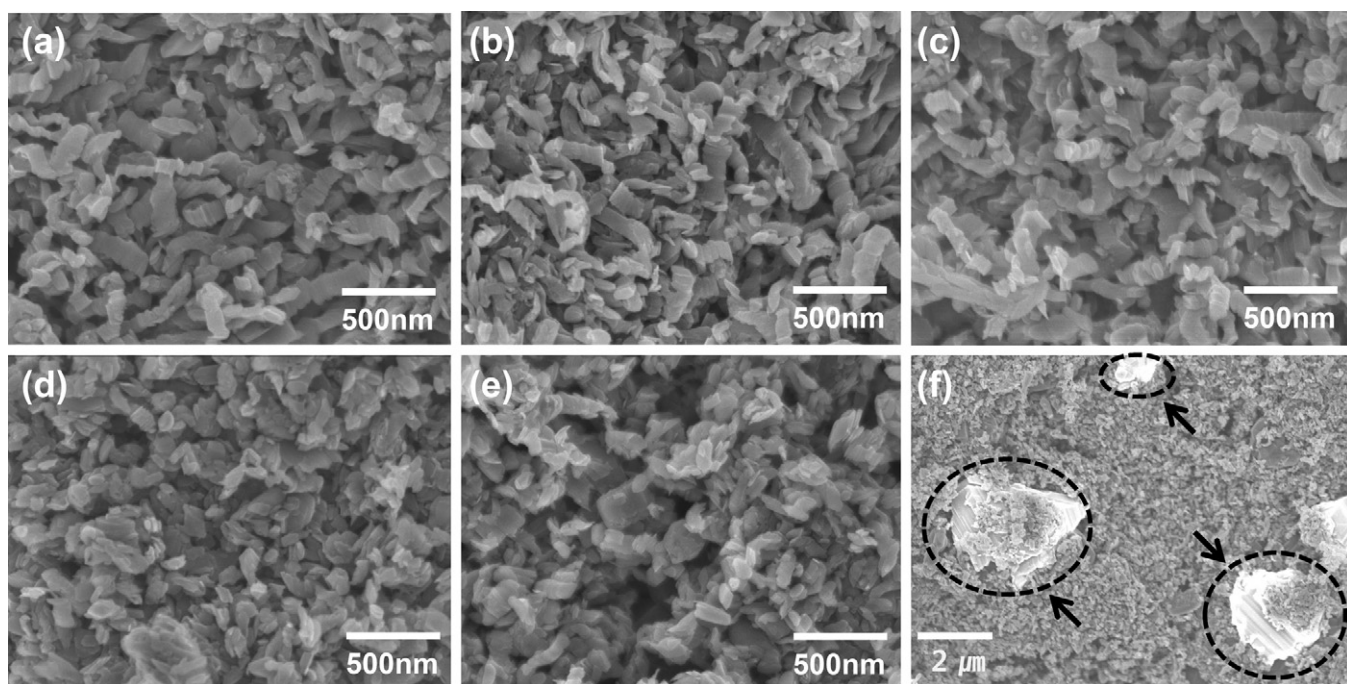


Fig. 2. SEM images of PCNF: (a) PCNF, (b) PCNF-G, (c) PCNF-G-NA, (d) PCNF-B-G, (e) PCNF-B-G-NA, and (f) B₄C particles dispersed in PCNF-B-G.

treatment decreased carbon content and increased the nitrogen and oxygen contents, suggesting introduction of nitrogen- and/or oxygen-containing functional groups to the edge surfaces of PCNF-G-NA. Boron doping decreased carbon content and increased the

O_{diff}/C ratio of boron-doped PCNF. The boron content was determined to be 3.4 wt% by elemental analysis. Fig. 1 shows the XRD profiles of various PCNFs, which showed a sharp peak around 26° . This indicated that PCNF had a highly graphitic structure.

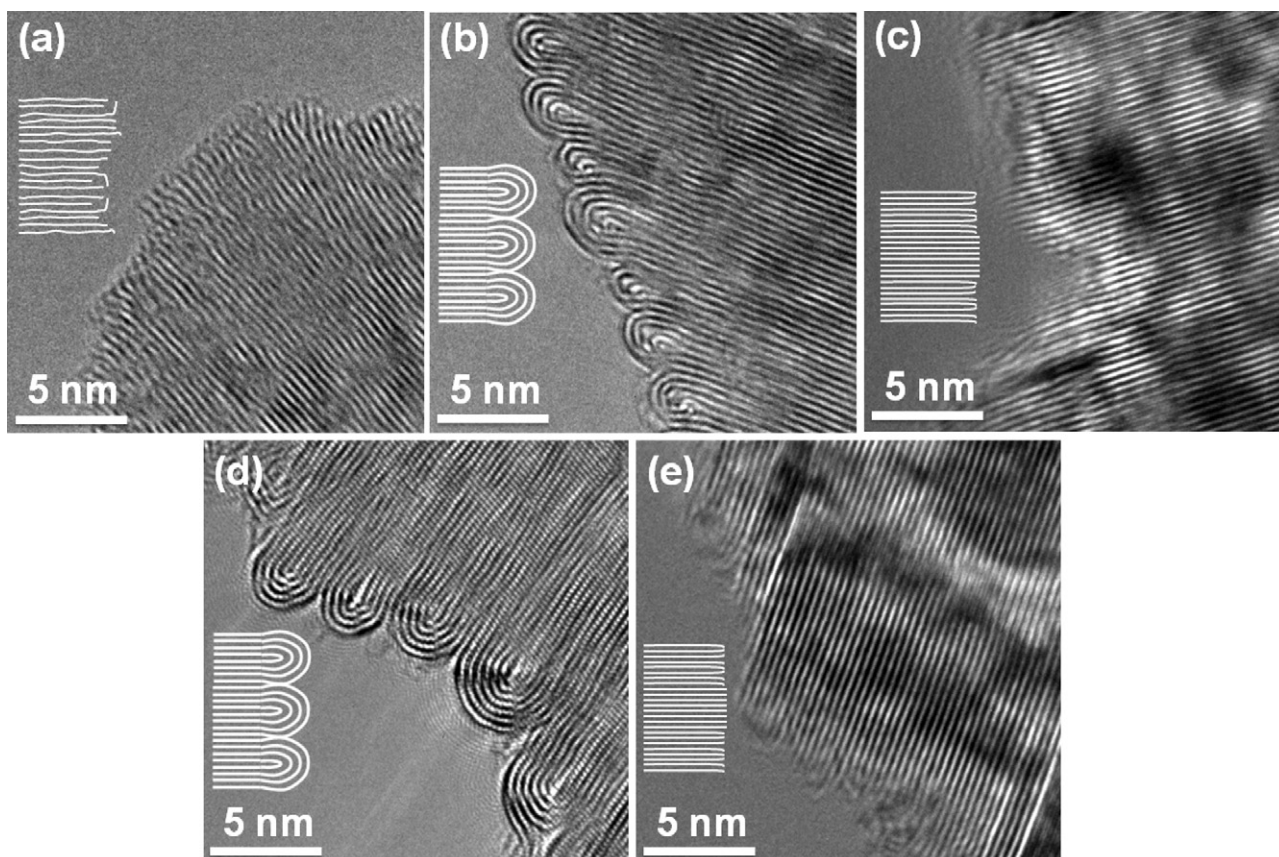


Fig. 3. TEM images of PCNF: (a) PCNF, (b) PCNF-G, (c) PCNF-G-NA, (d) PCNF-B-G, and (e) PCNF-B-G-NA.

Table 2

Charge, discharge, irreversible capacity, and coulombic efficiency of PCNF, PCNF-G, and PCNF-B-G at the first and second cycles.

Samples	Capacity at the first cycle (mAh g^{-1})				Capacity at the second cycle (mAh g^{-1})			
	Charge	Discharge	Irreversible	C.E. ^a (%)	Charge	Discharge	Irreversible	C.E. ^a (%)
PCNF	703	422	281	60.0	450	409	41	90.9
PCNF-G	595	376	219	63.2	393	365	28	92.9
PCNF-B-G	495	356	139	71.9	373	354	19	94.9

^a Coulombic efficiency (%) = (discharge capacity)/(charge capacity) $\times$ 100.

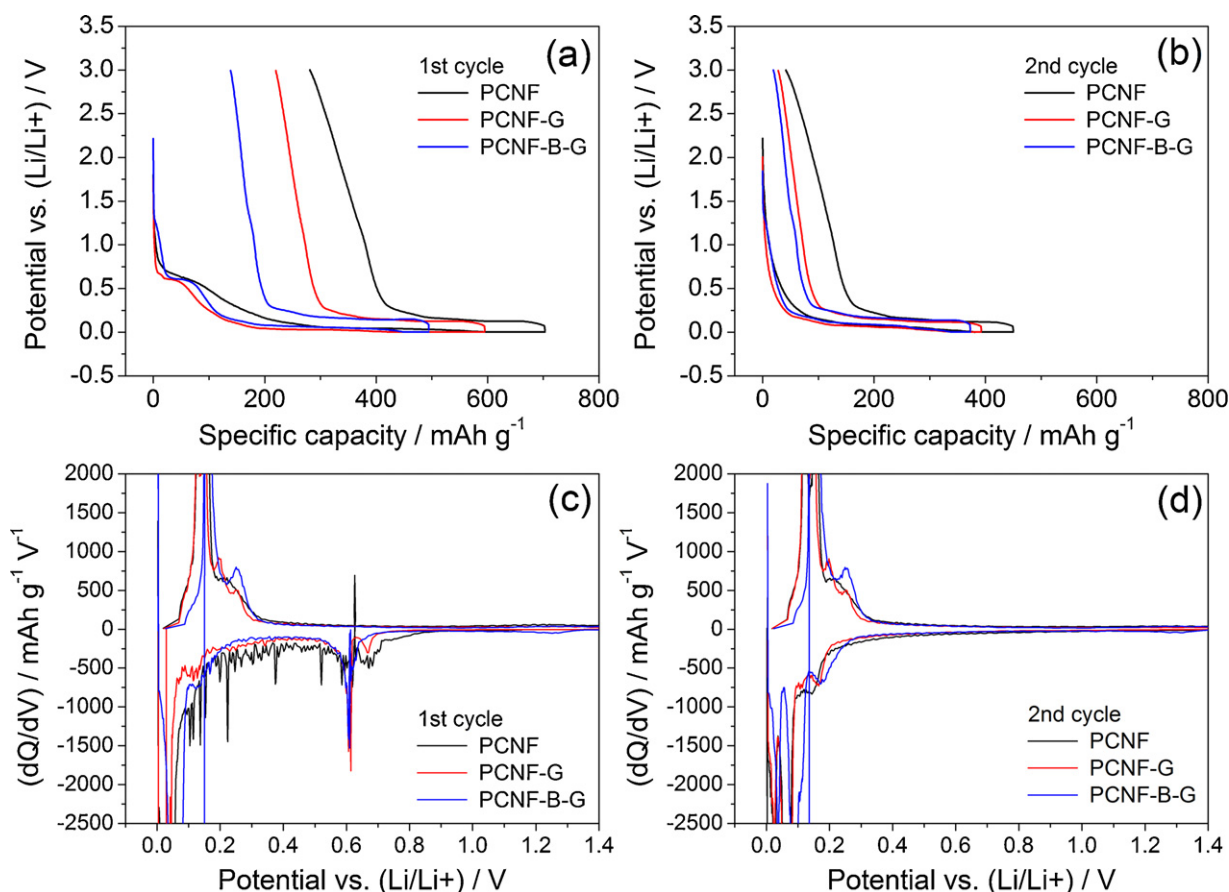
Table 1 shows the interlayer spacing (d_{002}) and stacking height (L_{c002}) of as-prepared, graphitized, nitric acid treated, and boron-doped PCNF. As-prepared PCNF showed a high degree of graphitization, with d_{002} and L_{c002} values of 3.365 Å and 57 nm, respectively. Boron-doped PCNF showed much higher degree of graphitization with smaller d -spacings, i.e., 3.359 Å, compared to those of non-boron-doped PCNF, 3.364–3.366 Å. The L_{c002} values of boron-doped PCNF were also higher than those of non-boron-doped PCNF. The L_{c002} values of PCNF-B-G and PCNF-B-G-NA were 99 nm and >100 nm, respectively. Boron doping of carbonaceous materials is known to enhance the degree of graphitization [13].

The surface area decreased due to heat treatment at 2800 °C and further decreased due to boron doping. As-prepared PCNF showed the highest R value because of its edge surfaces. Heat treatment at 2800 °C decreased the R value of as-prepared PCNF by closed loop formation on the surface. Nitric acid treatment increased the R value of PCNF-G due to newly exposed free edges on the surface by uniform cutting off of the loop ends. Lim et al. have suggested that the relaxation effect of surface tension by newly exposed free edges on the surface might increased slightly a degree of graphitization of nitric acid treated PCNF [10]. Boron doping increased the R

value even when PCNF-B-G had the same basal surfaces as PCNF-G. Hagio et al. showed that boron doping introduces local distortion around substitutional boron atoms within the graphite layer planes [14]. Usually, Raman spectroscopy evaluates the surface layer, of which depth is usually less than 20 nm in depth [15], while XRD is the most powerful method to evaluate total crystallinity. From the combinations of XRD and Raman spectroscopy, boron doping increased a degree of graphitization of PCNF, while it might induce local distortion and therefore amorphous structures on the surface of boron-doped PCNF.

3.2. Microscopic observation of PCNF

Fig. 2 shows SEM images of as-prepared, graphitized, nitric acid treated and boron-doped PCNF. The diameter of PCNF observed by SEM ranged from 60 to 160 nm. The length of graphitized PCNF decreased slightly due to graphitization compared to as-prepared PCNF. Graphitization decreased the length of PCNF because PCNFs are easily cleaved by heat treatment. The length of boron-doped PCNF decreased slightly due to ball milling for 24 h, as shown in Fig. 2(d) and (e). The boron content of PCNF-B-G was calculated

**Fig. 4.** Charge/discharge profiles (a and b) and differential capacity–voltage plots (c and d) of the first two cycles for PCNF, PCNF-G, and PCNF-B-G.

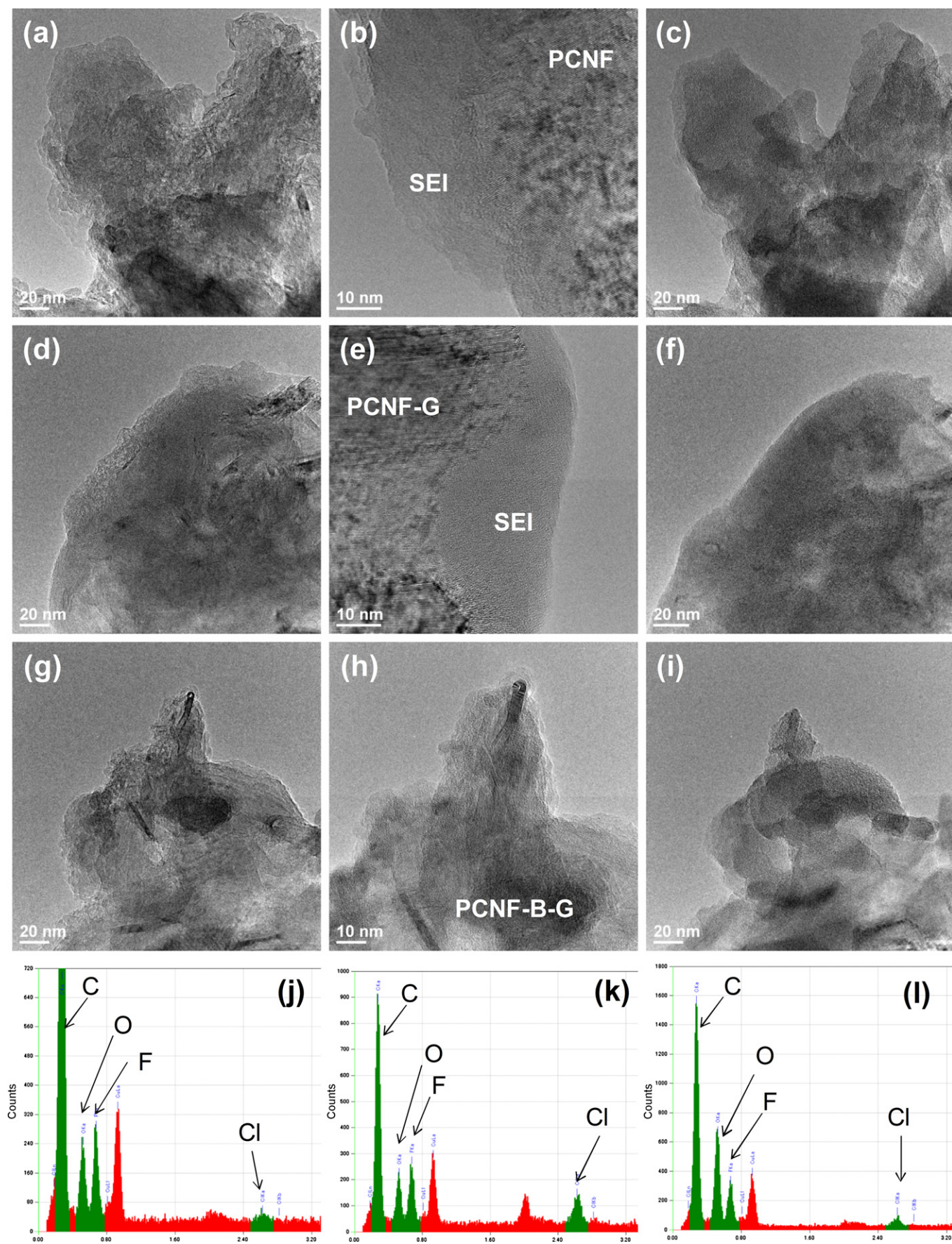


Fig. 5. TEM images of PCNF before EDS mapping (a and b), after EDS mapping (c), PCNF-G before EDS mapping (d and e), after EDS mapping (f), PCNF-B-G before EDS mapping (g and h), after EDS mapping (i), EDS spectra of PCNF (j), PCNF-G (k), and PCNF-B-G (l).

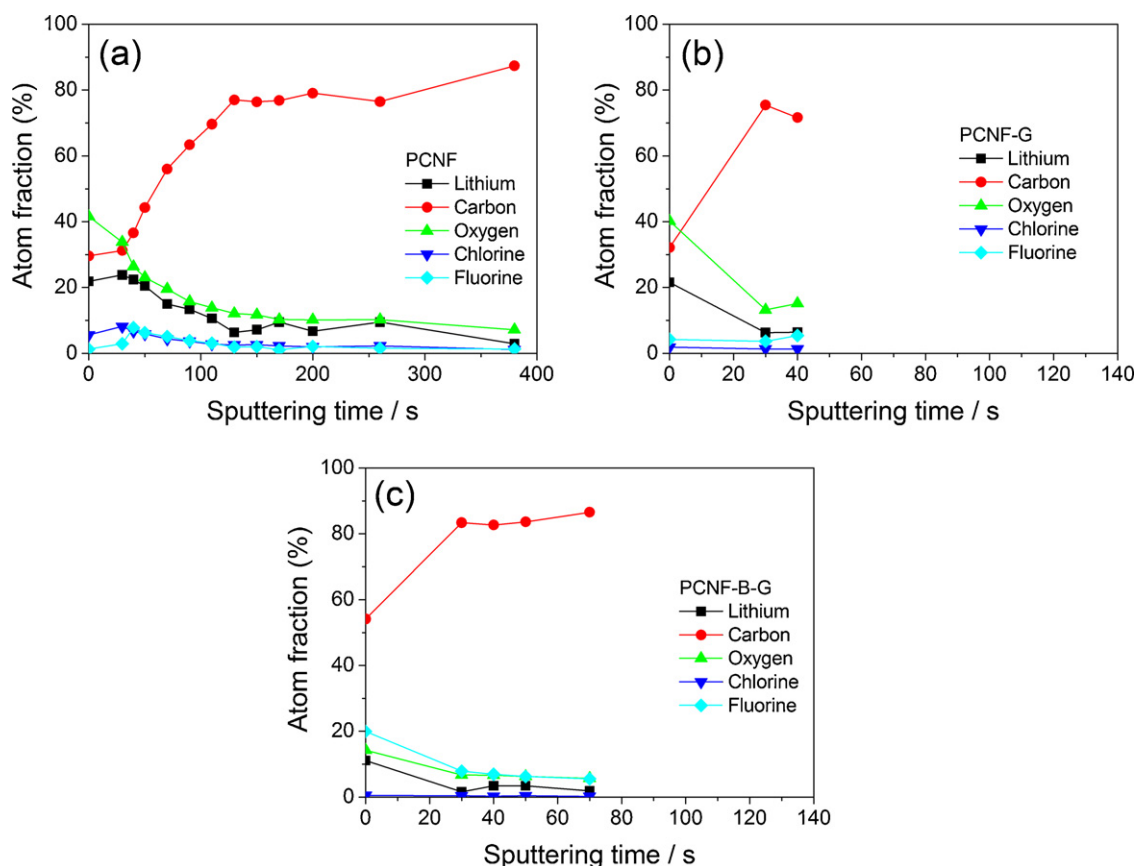


Fig. 6. XPS depth profiles of PCNF (a), PCNF-G (b), and PCNF-B-G (c) with Ar sputtering time.

as 3.4 wt% based on elemental analysis. Lowell reported that the maximum amount of substitutional boron in the graphite lattice is 2.35 at% (2.12 wt%) [16]. Excessive addition of boron results in the formation of boron carbide (B_4C) particles, about 1.0–5.0 μm in diameter, as shown by the arrowed marks in Fig. 2(f).

Fig. 3 shows TEM images of as-prepared, graphitized, nitric acid treated and boron-doped PCNF. Graphitic edges on the surface of as-prepared PCNF were arranged irregularly, which resulted in a high surface area. PCNF-G (b) and PCNF-G-NA (c) showed highly graphitized layers, unlike as-prepared PCNF (a). Graphene layers of PCNF-G were arranged in parallel, unlike as-prepared PCNF. Graphitization induced closed loop ends on the surface, as shown in Fig. 3(b) and (d), and thus its surface likely consisted of curved basal planes. Nitric acid treatment cut off the closed loop ends, exposing free edges, as shown in Fig. 3(c) and (e) [10].

3.3. Effects of edge and basal surfaces on the SEI formation

Table 2 summarizes the charge, discharge, irreversible capacity, and coulombic efficiency of PCNF, PCNF-G, and PCNF-B-G in the first and second cycles. The irreversible capacity loss at the first cycle is almost given by the capacity needed for the formation of the SEI. In the second cycle and beyond, the irreversible capacity decreased remarkably because the SEI formation reaction is limited to the first cycle.

The irreversible capacity of PCNF with edge surfaces was higher than those of PCNF-G and PCNF-B-G with basal surfaces in the first cycle. The high irreversible capacity of PCNF was probably due to abundant active edge planes on the surfaces, whereas the low irreversible capacities of PCNF-G and PCNF-B-G were probably related to inactivity of the curved basal planes composed of closed loop

ends on the surfaces. Edge planes were more favorable at charging the surfaces of the carbon materials by the well-known anisotropic conductivity because the π -electrons moved more easily along the graphitic planes to the edges under electrochemical polarization [9]. Reactivity to electrolytes at edge surfaces is much higher than that at basal surfaces and can easily result in decomposition of the electrolyte solution during the charge process in the first cycle, leading to a lower coulombic efficiency. As a result, the coulombic efficiency of PCNF with edge surfaces was lower than those of PCNF-G and PCNF-B-G with basal surfaces. The coulombic efficiencies of PCNF, PCNF-G, and PCNF-B-G were 60.0, 63.2, and 71.9%, respectively. The low discharge capacity of PCNF-G may

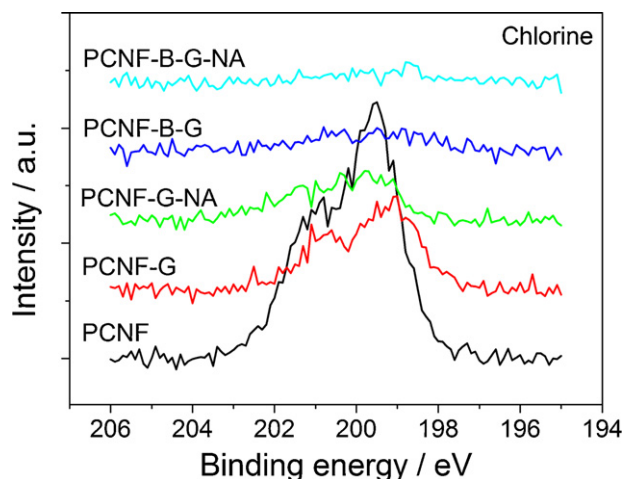


Fig. 7. XPS spectra of chlorine in the CNF before Ar sputtering.

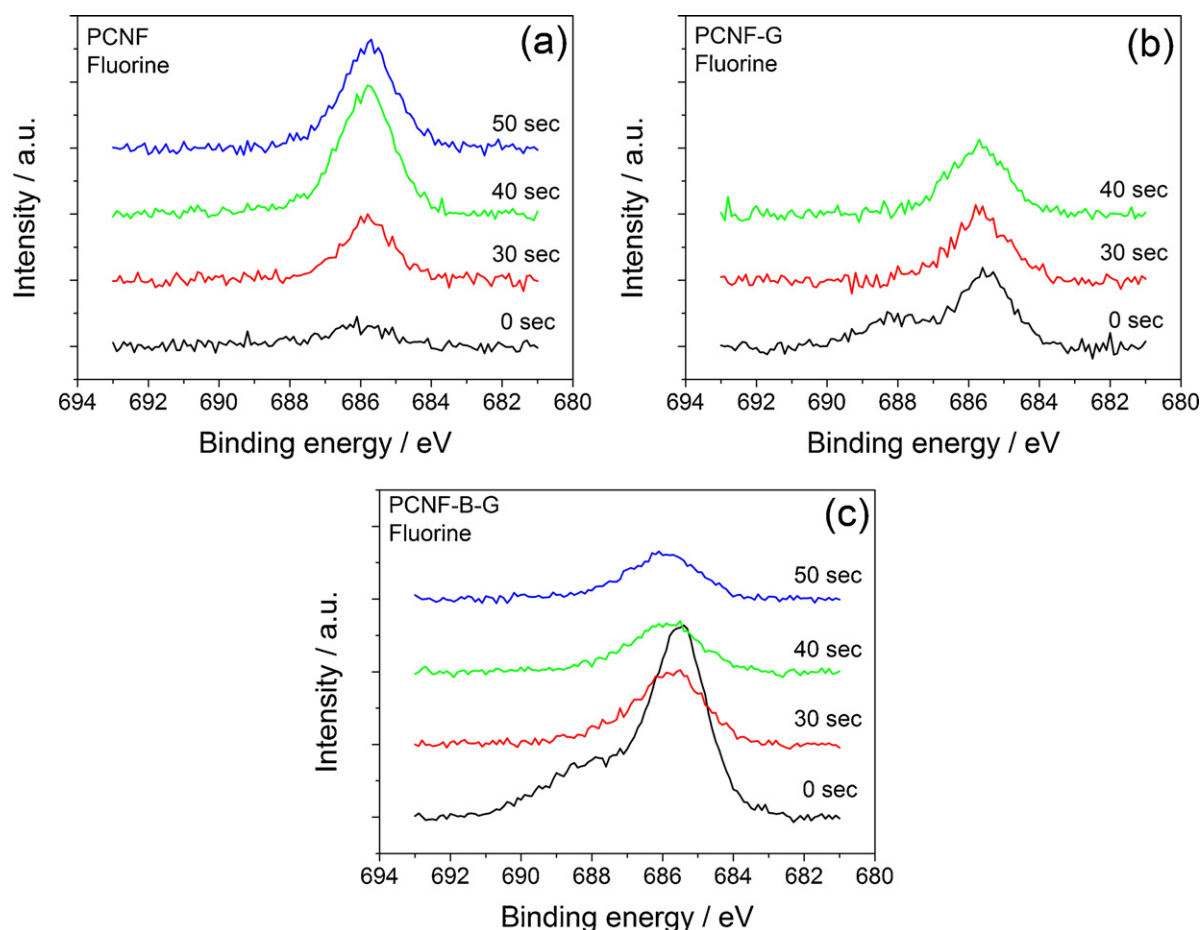


Fig. 8. XPS spectra of fluorine in PCNF (a), PCNF-G (b), and PCNF-B-G (c) with Ar sputtering time.

be associated with the presence of the basal planes from the loop formation. Habazaki et al. suggested that loops may impede the intercalation of lithium ions into all the inter-layers in PCNF-G [17].

Fig. 4 shows the charge/discharge profiles and differential capacity–voltage plots of the first two cycles for PCNF with edge surfaces, PCNF-G, and PCNF-B-G with basal surfaces that were charged and discharged at a current density of 372 mA g^{-1} (1 C-rate) between 0.003 and 3.0 V vs. Li/Li⁺. The irreversible capacity was observed from the charge/discharge profiles and differential capacity–voltage plots and was associated with SEI formation on the surfaces of PCNF by the decomposition of the electrolyte and lithium salt. Fig. 4(c) shows that the cathodic reaction of PCNF, PCNF-G, and PCNF-B-G initially appeared at about 0.87, 0.75, and 0.77 V, respectively, and the cathodic peak of PCNF and PCNF-G was observed at around 0.6 V. Even below 0.6 V, the cathodic reaction of PCNF with edge surfaces was larger than those of PCNF-G and PCNF-B-G with basal surfaces. Fig. 4(d) shows that the cathodic peaks of PCNF and PCNF-G disappeared at the second cycle.

Fig. 5 shows the morphology and energy dispersive X-ray spectra of the SEI on the surfaces of PCNF, PCNF-G, and PCNF-B-G by

TEM-EDS. SEI morphology was observed after five charge/discharge cycles between 0.003 and 3 V to stabilize SEI formation. The surfaces of PCNF, PCNF-G, and PCNF-B-G were observed to be covered with a thin sponge layer composed of amorphous or less-crystalline reduction products in the TEM images of Fig. 5. The SEI on the surface of PCNF was observed to be more abundant than those of PCNF-G and PCNF-B-G. Carbon, chlorine, oxygen, and fluorine peaks were detected in the EDS spectra of PCNF, PCNF-G, and PCNF-B-G, as shown in Fig. 5(j), (k), and (l) before mapping. The SEI formed on the surface of each CNF disappeared after EDS mapping because of continuous electron beam irradiation, as shown in Fig. 5(c), (f), and (i).

Fig. 6 shows the depth-dependent elemental composition of PCNF, PCNF-G, and PCNF-B-G with Ar sputtering time. Eshkenazi et al. suggested that the sputtering efficiency would not be markedly altered by the type of material, such as organic materials and stable inorganic materials, material decomposition, and surface chemical reactions, as confirmed by TOF-SIMS and FT-IR analyses [7]. The concentration of oxygen formed by a reduction of electrolyte decreased continuously with the sputtering time in

Table 3

Charge, discharge, irreversible capacity, and coulombic efficiency of PCNF-G-NA and PCNF-B-G-NA at the first and second cycles.

Samples	Capacity at the first cycle (mAh g^{-1})				Capacity at the second cycle (mAh g^{-1})			
	Charge	Discharge	Irreversible	C.E. ^a (%)	Charge	Discharge	Irreversible	C.E. ^a (%)
PCNF-G-NA	646	412	234	63.8	435	401	34	92.2
PCNF-B-G-NA	496	370	126	74.6	392	371	21	94.6

^a Coulombic efficiency (%) = (discharge capacity)/(charge capacity) × 100.

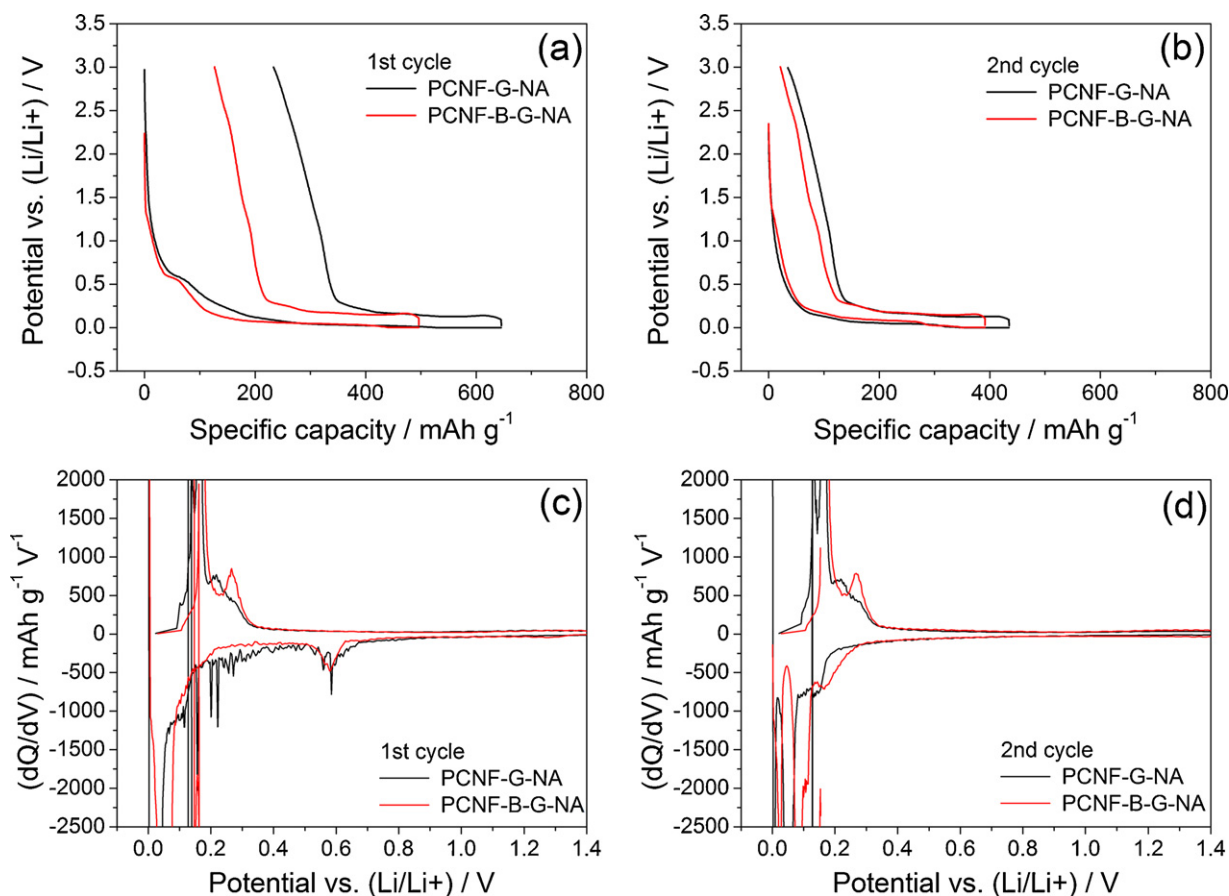


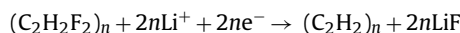
Fig. 9. Charge/discharge profiles (a and b) and differential capacity–voltage plots (c and d) of the first two cycles for PCNF-G-NA and PCNF-B-G-NA.

PCNF, PCNF-G, and PCNF-B-G, being higher at the electrolyte side of the SEI. The continuous increase in carbon concentration with sputtering time can be explained by the signal from carbon of PCNFs underlying the SEI. The concentration of carbon in PCNF, PCNF-G, and PCNF-B-G was almost saturated after 130, 30 and 30 s of sputtering time, respectively. After 30 s of sputtering time, the total atomic concentration of Li, F, Cl and O in PCNF, PCNF-G, and PCNF-B-G were 68.7%, 24.6%, and 16.6%, respectively, and almost saturated in PCNF-G and PCNF-B-G. The SEI of PCNF with edge surfaces was much more abundant and four times thicker than those of PCNF-G and PCNF-B-G with basal surfaces. The XPS depth profiles of PCNF, PCNF-G, and PCNF-B-G were consistent with the irreversible capacity results.

The maximum atomic concentration of chlorine in PCNF, PCNF-G and PCNF-B-G was 8.2%, 1.9% and 0.5%, respectively. This means that the ClO_4^- anion is preferentially reduced on the edge surfaces of PCNF. The chlorine peak (Cl_{2p} peak) assigned to LiCl at 199.0–200.0 eV was observed in the XPS spectra (Fig. 7) before Ar sputtering. Many researchers have reported that the chlorine peak assigned to LiCl was observed at 198.4 eV [18], 199 eV [19,20], and 200 eV [21] in the XPS spectra.

The fluorine peak of the TEM-EDS spectrum and XPS spectra was mainly due to LiF because the PVDF binder was rapidly evaporated by the electron beam. The fluorine peak assigned to LiF at 685.5–686 eV was observed in the XPS spectra (Fig. 8), whereas the fluorine peak assigned to CF_2 bonds at 689 eV was scarcely observed with Ar sputtering. Aurbach et al. showed that the PVDF binder in a 1 M $\text{LiClO}_4/\text{EC-DMC}$ (1:1) electrolyte was partly reduced at low potentials, and C–F bonds were broken, forming LiF and possibly carbide species C–Li [21]. Liu et al. have suggested that PVDF binder

could be reduced electrochemically to LiF in Li/PVDF-mixed Ni cell according to the following reaction [22]:



Fransson et al. have reported that a reduction reaction of PVDF binder was observed at 0.35 V vs. Li/Li^+ in carbon black electrodes [23].

3.4. Effects of boron doping on the SEI formation

Table 3 summarizes the charge, discharge, irreversible capacity, and coulombic efficiency of PCNF-G-NA and PCNF-B-G-NA at the first and second cycles. The irreversible capacity of PCNF-G-NA was higher than that of PCNF-B-G-NA. The low irreversible capacity of boron-doped carbon materials may be related to the redistribution of the Fermi level of the boron-doped carbon materials, which is lowered by boron-doping by introducing an electron acceptor into the lattice [24]. Boron is known to enhance the graphitization of non-graphitic carbonaceous materials, such as pitch and pitch-based carbon fibers, by substitutional incorporation into the graphene layers and defects, enhancing the lithium intercalation capacity. The charge and discharge capacities of PCNF-B-G-NA were lower than those of PCNF-G-NA even though the degree of graphitization in PCNF was slightly enhanced by boron doping. This was due to the formation of boron carbide (B_4C), as shown by SEM, which did not contribute to the electrochemical reaction [8].

Fig. 9 shows the charge/discharge profiles and differential capacity–voltage plots of the first two cycles for PCNF-G-NA and PCNF-B-G-NA with the same edge surfaces. Fig. 9(c) shows that

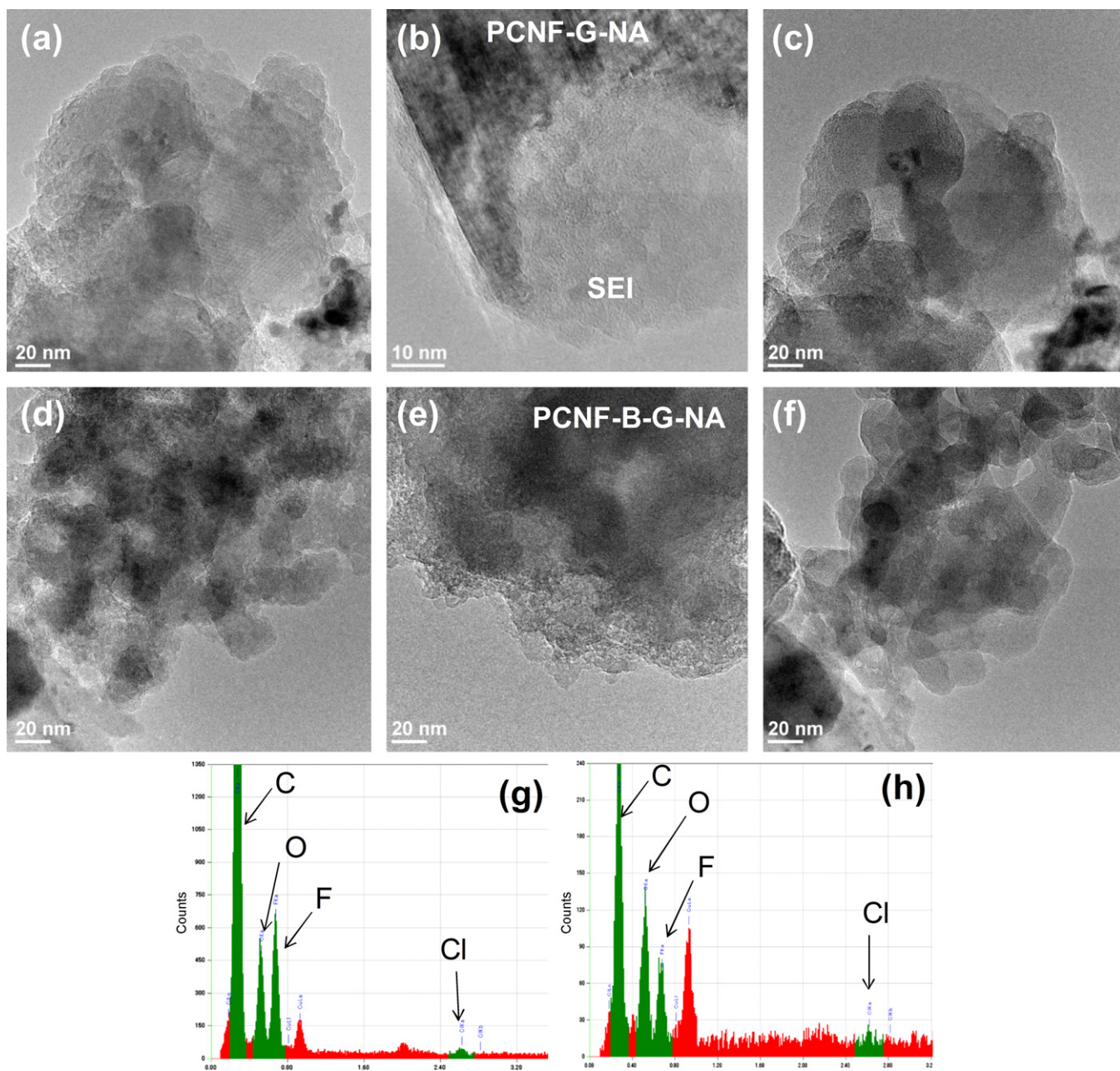


Fig. 10. TEM images of PCNF-G-NA before EDS mapping (a and b), after EDS mapping (c), PCNF-B-G-NA before EDS mapping (d and e), after EDS mapping (f), EDS spectra of PCNF-G-NA (g), and PCNF-B-G-NA (h).

the cathodic reaction of PCNF-G-NA and PCNF-B-G-NA initially appeared at 1.20 and 1.13 V, respectively, and the cathodic peaks of PCNF-G-NA and PCNF-B-G-NA were observed at about 0.6 V, similar to PCNF, PCNF-G, and PCNF-B-G. Below 0.6 V, the cathodic reaction of PCNF-G-NA was larger than that of PCNF-B-G-NA. The electrochemical Li^+ intercalation/deintercalation took place at a higher voltage of 50 mV in PCNF-B-G-NA than in PCNF-G-NA, as shown in Fig. 9(d). Endo et al. reported that the voltage of boron-doped milled mesophase pitch-based carbon fibers (mMPCF) was 40 mV higher for Li^+ intercalation/deintercalation than non-doped mMPCF, presumably because the substitutional boron acted as an electron acceptor in the graphite lattice and affected the exposed dislocation edge-type surface [25].

Fig. 10 shows the morphology and energy dispersive X-ray spectra of the SEI on the surfaces of PCNF-G-NA and PCNF-B-G-NA

by TEM-EDS. The surfaces of PCNF-G-NA and PCNF-B-G-NA were observed to be covered with a thin layer of small reduction products, such as PCNF and PCNF-G. The SEI morphology of PCNF-G-NA is a mosaic type, which is composed of various less-crystalline or amorphous reduction products (Fig. 10(b)). The SEI on the surface of PCNF-G-NA was observed to be more abundant than that of PCNF-B-G-NA. Carbon, chlorine, oxygen, and fluorine peaks were detected in the EDS spectra of PCNF-G-NA and PCNF-B-G-NA, as shown in Fig. 10(g) and (h) before mapping.

Fig. 11 shows the depth-dependent elemental compositions of PCNF-G-NA and PCNF-B-G-NA with Ar sputtering time. The concentration of carbon increased with sputtering time due to carbon of PCNFs underlying the SEI, being almost saturated after 110 s and 30 s of sputtering time, respectively. The SEI of PCNF-G-NA was much more abundant and three times thicker than

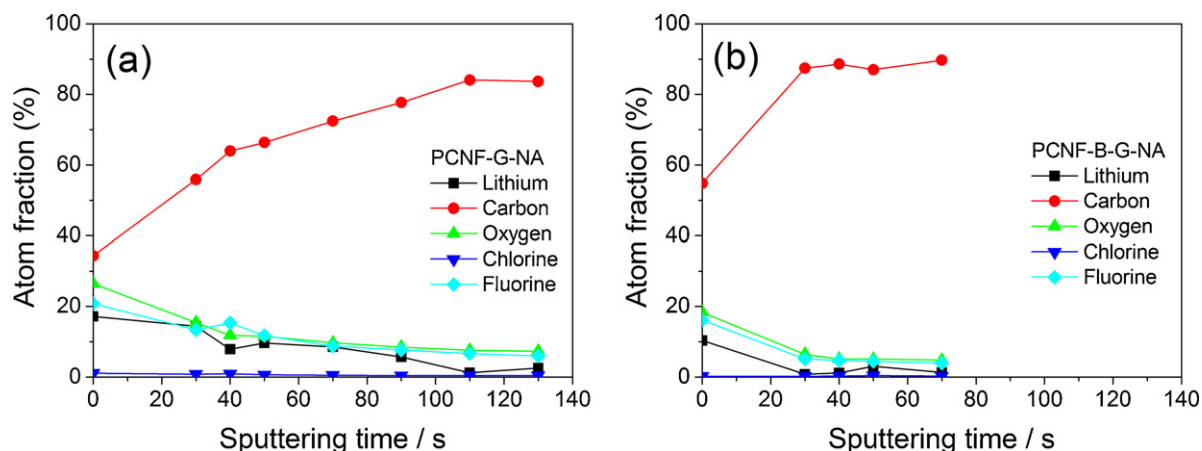


Fig. 11. XPS depth profiles of PCNF-G-NA (a) and PCNF-B-G-NA (b).

that of PCNF-B-G-NA. The XPS depth profiles of PCNF and PCNF-G were consistent with the irreversible capacity results. The fluorine peak of the EDS spectrum was primarily due to LiF rather than the PVDF binder as shown in the fluorine peak of XPS spectra in PCNF, PCNF-G, and PCNF-B-G. Before Ar sputtering, the total atomic concentration of Li, F, Cl and O in PCNF-G-NA and PCNF-B-G-NA was 65.6% and 45.1%, respectively. After 30 s of sputtering time, the concentration of Li, F, Cl and O in PCNF-G-NA and PCNF-B-G-NA was 44.1% and 12.6%, respectively, and almost saturated in PCNF-B-G-NA. It could be concluded that boron doping inhibited the formation of SEI in boron-doped PCNF and therefore decreased the irreversible capacity at the first cycle even though the surface of PCNFs was composed of identical edge planes.

4. Conclusions

The irreversible capacity loss of PCNF (281 mAh g^{-1}) with edge surfaces was higher than those of PCNF-G (219 mAh g^{-1}) and PCNF-B-G (139 mAh g^{-1}) with basal surfaces. The XPS depth profiles indicated that the SEI of PCNF with edge surfaces was more abundant and four times thicker than those of PCNF-G and PCNF-B-G with basal surfaces. The surface of well-defined PCNF was covered with a thin sponge layer composed of amorphous or less-crystalline reduction products. Boron doping decreased the irreversible capacity loss and SEI formation. The irreversible capacity loss of boron-doped PCNF-B-G-NA (126 mAh g^{-1}) with edge surfaces was lower than that of non boron-doped PCNF-G-NA (234 mAh g^{-1}) with edge surfaces. The SEI of PCNF-G-NA without boron doping was much more abundant and three times thicker than that of PCNF-B-G-NA with boron doping. It could be concluded that the surfaces with basal planes and boron doping are the most effective way to minimize the irreversible capacity loss and SEI formation from results of PCNFs with well-defined surfaces.

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